# FUNCTIONAL REPORT

AI-powered educational application — student experience and teacher tools

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**Product: Magicbook / educational application**
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**Basis of analysis: screenshots provided by the user**

## Document purpose

To provide a structured description of the application's observable features, user journeys, data involved, AI interactions, inferred business rules, and the points that need clarification or validation during specification and implementation.

1. Executive summary

The application is positioned as an AI-powered learning environment that transforms handwritten notes and study materials into organised digital content and learning activities. The main observed workflow is: the student scans a page, reviews the extracted text, links it to a subject and chapter, and can then view a digitised version, request explanations, generate study materials, or interact with the AiA assistant.

The experience also includes standalone tools — flashcards, quizzes, mind maps, a semantic checker, and an essay checker —, organisation by subjects and chapters, day-based study paths, and a class area with presentations and assessments. The screenshots also show an onboarding flow that collects the student's preferences and difficulties to personalise the experience.

| Area | Main capability | Role in the product |
|---|---|---|
| Onboarding | Collection of profile, country, preferences, and difficulties | Personalise content and recommendations |
| Scanning/OCR | Capture, recognition, and review of notes | Convert physical content into usable content |
| Organisation | Subjects, chapters, and pages | Create context and a knowledge structure |
| AI study tools | Explain, summarise, and generate flashcards, quizzes, and mind maps | Support understanding, revision, and practice |
| AiA | Context-aware conversational assistant | Guide learning based on the student's notes |
| Assessment | Quizzes with feedback and essay checking | Measure performance and provide guidance |
| Teacher/class | Classes, presentations, and assessments | Provide materials and activities to students |

| | | |
|---|---|---|
| Learning path | Sequence of activities with statuses and locked stages | Guide progress throughout the learning process |

2. Scope, assumptions and limitations of the analysis

This report describes only what is visible in the screenshots and organises the functional workflows that can reasonably be inferred from them.

It is not assumed that all screens or features are available in production, across all platforms, or for every user profile.

The teacher area is represented by a list of classes, presentations, and assessments; insufficient screenshots were provided to specify class creation/editing, student management, publishing, or results analysis.

The onboarding flow and some screens are in English, while others are in Portuguese. The application should have a consistent localisation strategy.

Access rules, usage limits, pricing, permissions, data retention, and external integrations cannot be confirmed from the screenshots alone.

Convention: when a rule is not explicitly demonstrated, it is identified as a recommendation or an item to be confirmed, rather than as functionality that has already been implemented.

3. User profiles

3.1 Student

The primary user, who scans notes, organises subjects, studies with AI support, completes quizzes, and tracks learning progress.

3.2 Teacher

A user who manages classes and provides presentations and assessments. The screenshots confirm the display of classes, subjects/grade levels, presentations with slide counts, and assessments with a start action.

3.3 Academic context

The application links content to subjects and chapters and shows examples of education levels and curriculum contexts from Portugal and Brazil. The country and educational pathway should influence personalisation, although the full rule set is not visible.

4. Student onboarding and personalisation

The onboarding flow progressively collects information, using a progress bar and questions presented by the AiA mascot.

| Observed step | Data/options | Functional purpose |
|---|---|---|
| Country | Portugal (editable) | Select the curriculum, language, and educational context |
| How do you learn best? | Visual, auditory, reading and writing, hands-on practice | Adapt formats and study suggestions |
| What matters most? | Improve grades, study faster, catch up | Identify the main goal |
| What is holding you back at school? | Procrastination, exam stress, memory, understanding, knowing what to study | Identify difficulties and personalise support |
| Personal data | Full name and date of birth | Personalisation, identification, and age-appropriate adaptation |

Functional requirements

Allow users to select one or multiple preferences when the screen presents multiple options.

Allow users to edit the selected country and validate required information before completion.

Save the answers to the student's profile and make them available to the personalisation engine.

Use the date of birth to calculate the user's age without unnecessarily exposing this information in the interface.

Allow users to review or change their preferences later in their profile settings.

Items to validate

Determine whether learning preferences are treated as self-reported preferences, rather than as a diagnosis or fixed classification of the student.

Define consent, parental information, and the processing of minors' data in accordance with the relevant market and applicable legislation.

Ensure that country, interface language, content language, and school curriculum are treated as separate fields where necessary.

5. Navigation and main structure

The observed bottom navigation includes five entries:

| Entry | Observed/inferred function |
|---|---|
| Home | Access to the home area and a summary of the student's work. |
| Subjects | Subject list and organisation; includes controls to expand/collapse and reorder. |
| Central scan button | Main action to start capturing a page. |
| AiA | Access to the conversational learning assistant. |
| Menu | Access to other areas and settings; content not visible in the screenshots. |

The subject list includes examples such as Geography, Psychology, Chemistry, Biology, Law, Literature, Mathematics, and Medicine. A “+” button is available to add subjects.

Functional requirements

Create, edit, add, reorder, and expand/collapse subjects.

Each subject should allow chapters and scanned pages to be linked to it.

Keep navigation consistent across Home, Subjects, scanning, AiA, and Menu.

Define empty states for users who have no subjects or content.

6. Smart scanning and OCR

Scanning is the product's main entry point. The welcome screen offers three free scans using any page from any notebook so users can try the application. After this limit, the displayed message indicates unlimited use with compatible Jandaia notebooks.

6.1 Observed functional workflow

The student starts scanning using the central button or the “Scan now” action.

The application opens the camera and displays visual guidance for framing the page.

The page is captured and the system displays a processing state (“Reading your page…”).

OCR extracts the handwritten and/or printed content.

The user is taken to a review screen (“Review the text”), where they can check and correct the extracted content.

After confirming, the student links the page to a subject and a chapter.

The page then becomes available within the subject/chapter context.

6.2 Review of extracted text

The review screen displays the recognised text in an editable field and explains that the user should check and correct the content before proceeding.

Allow OCR errors to be corrected before generating any derived content.

Preserve paragraphs, headings, lists, numbering, symbols, and hierarchy where recognised.

Save the corrected version as the primary source for subsequent features.

Allow users to cancel or go back without losing the capture, requesting confirmation when changes have been made.

6.3 Linking to a subject and chapter

The “Save to your notebook” screen displays the page image and two fields: Subject and Chapter. The user confirms the association.

Allow users to select an existing subject and chapter.

Allow users to create a new chapter if this is part of the product workflow.

Allow users to change the association later.

Save the original image, OCR text, reviewed version, and page metadata.

6.4 Displayed access rules

| Observed rule | Displayed behaviour |
|---|---|
| Trial use | 3 free scans using any page/notebook. |
| After the 3 scans | Unlimited use indicated for compatible Jandaia notebooks. |
| Alternative action | “Not now” allows the user to postpone scanning. |

Notebook eligibility, brand identification, and scan counting should be validated by the backend to prevent discrepancies between what the interface communicates and what the service authorises.

7. Actions available after scanning

After the page has been processed, the “What would you like to do with this page?” screen is displayed, with four main options:

| Action | Functional description | Expected result |
|---|---|---|
| View scanned content | Opens the OCR content organised within the chapter. | Searchable text linked to the subject. |
| Explain it to me | The tutor explains the content step by step. | A pedagogical explanation grounded in the source page. |
| Generate content | Transforms notes into complete content. | Structured study material generated from the notes. |
| Summarise | Creates a quick overview. | A summary faithful to the notes. |

The scanned-content screen displays the subject, chapter, a “Back to original page” action, a copy control, and options such as “Use AI”, “Listen”, and “Voice note”, as well as navigation between pages in the chapter.

Cross-functional requirements

All generated responses must remain linked to the source page and text.

The AI must distinguish information explicitly present in the notes from supplementary information.

When the image or OCR output is insufficient, the system must indicate uncertainty instead of inventing content.

The listen option should provide text-to-speech; the voice note option should allow users to record spoken information, if this capability is enabled.

Navigation between pages must follow the chapter order and indicate the current position.

8. AiA learning assistant

AiA is presented as a conversational assistant that understands the student's context. In the example, it greets the student by name, says it has read their Geography notes, and asks what they would like to work on.

8.1 Quick suggestions

Create a quiz about the subject or topic.

Create flashcards.

Explain a concept.

Summarise the subject matter.

Suggest what might appear in the exam.

8.2 Open-ended interaction

The “Ask for anything” input allows users to type a question, use the microphone, and send a message. The “+” button suggests additional actions or attachments, although the details are not visible.

8.3 Persistent quick actions

Shortcuts for Quiz, Flashcards, and Mind Map are displayed, with horizontal scrolling available to access other tools.

Functional and AI requirements

Use the selected subject, chapter, and pages as the conversation context.

Allow the student to explicitly specify another subject or ask questions without preselected context.

Provide quick suggestions that are relevant to the available content.

Maintain conversation history and allow users to resume a session.

Support text and voice input, with clear recording, transcription, sending, and error states.

Generate responses appropriate to the student's education level and language/curriculum.

When a student asks for exam predictions, present them as revision suggestions, never as guarantees about what will be assessed.

9. AI-generated study tools

9.1 Flashcards

Flashcards are displayed as cards containing an illustrative image, a question/concept, and an instruction to tap to reveal the answer. Progress is shown (e.g. “Card 1 of 10”) along with a percentage bar.

Generate a set of cards from a page, chapter, subject, or topic.

Display one question or concept at a time and reveal the answer through user interaction.

Allow users to move forward/backward and track progress.

Save the set and allow users to resume it.

Validate answer accuracy and the quality of generated or selected images.

9.2 Quiz

The quiz displays one question at a time, answer options, numerical and percentage progress, a confirmation button, and an option to exit/delete the session.

Display a readable question and clearly labelled answer options (A, B, C, D).

Allow users to select an option and submit their answer.

Immediately display correct/incorrect feedback, the correct answer, and an explanation.

Move to the next question and calculate the final result.

Save attempts and performance if activity history is part of the product.

Critical quality observation

In one screenshot, option A is highlighted in green and marked with a check, but the panel below displays “Incorrect” while also stating “Almost, the correct answer is A”. This combination is contradictory. It should be investigated as a possible grading-logic error, a mismatch between the selected answer and the answer key, or an incorrect visual feedback state. Testing the consistency between the selected option, correct answer, colour/icon, and explanatory text is a priority.

9.3 Mind maps

The application generates mind maps from the student's notes. The European Union Law example shows a central node with branches for the European Union in numbers, curriculum, bibliography, key

legislation, and online resources.
Generate a hierarchical structure of concepts and sub-concepts.
Maintain coherent semantic relationships and preserve the source topics.
Allow users to view the map in full screen, zoom in/out, and pan.
Allow users to regenerate and save the map, and ideally export or share it.
Avoid excessive text in nodes and use readable hierarchy levels.

10. Writing checkers and assessment

10.1 Essay checker

The observed workflow allows the user to select a topic and submit an essay by taking a photo or uploading a PDF. The next screen displays a processing state ("Correcting your essay…").
The displayed result includes a title, word and paragraph counts, an overall score (example: 940), general feedback, strengths, and text analysis.
Allow users to select or confirm the essay topic.
Accept photos and PDFs; validate format, quality, and legibility.
Extract handwritten or digital text and allow recognition errors to be identified.
Assess the essay using a configurable rubric appropriate to the selected exam/market.
Display the score, general feedback, strengths, areas for improvement, and practical recommendations.
Identify the assessment criteria and explain how each criterion contributes to the score.
Do not present a score as official if it is an AI-generated estimate.

10.2 Semantic checker

The screen displays a text containing identified issues and highlighted passages. The user is instructed to tap a highlighted passage to understand why it was flagged.
Identify potentially problematic passages in study texts.
Highlight the passage within its original context.
When tapped, explain why the passage was flagged and suggest a rewording or correction.
Distinguish between factual errors, inaccuracies, contradictions, ambiguity, redundancy, and language issues.
Allow users to accept, reject, or edit suggestions.
Item to validate: determine whether the semantic checker is intended for notes, answers, academic texts, or all these content types, and define the analysis criteria for each case.

11. Academic organisation: subjects, chapters, and pages

The application organises information by subject and chapter. The subjects area allows users to add and reorder subjects. Scanned content is linked to a chapter and can be viewed as individual pages.

| Entity | Suggested functional fields/concepts |
|---|---|
| Subject | Name, icon, optional colour, display order, education level/grade. |
| Chapter | Name, linked subject, order, content, and status. |
| Page | Original image, OCR text, reviewed text, capture date, chapter, source. |
| Generated resource | Type (summary, quiz, flashcards, mind map), source, date, version. |

It is recommended that every generated resource retain references to its source material, allowing students to return to the notes from which the content was created.

12. Learning path and progress

One screenshot shows a visual day-by-day learning path associated with a topic ("Evolution and natural selection"). Day 2 shows completed Flashcards and a Quiz as the next activity; Day 3 shows a locked

activity.

Represent activities in a learning sequence organised by day or stage.

Display distinct states: completed, next, available, locked, and not started.

Visually highlight the next activity.

Link each activity to a subject and topic.

Define unlocking conditions, such as completion of the previous activity or an availability date.

Allow users to resume the learning path and preserve their progress.

Item to confirm: whether the learning path is generated automatically based on the difficulties and goals identified during onboarding, created by the teacher, or configured by the student.

13. Classes, presentations, and assessments area

The "My Classes" area displays a list of classes with their name, subject, grade level, and the label "Powered by Magicboard". A "+" button is available to add a class.

13.1 Class content

When a class is opened, the interface displays two sections:

Presentations: cards with a thumbnail, title, slide count, and access to the content.

Assessments: assessment activities with a title and a "Start" action.

13.2 Requirements to be specified

Define teacher and student roles and permissions.

Specify how a class is created, how students join, and how they are linked to a subject/grade level.

Define how presentations are published, updated, and removed.

Define whether assessments have deadlines, attempt limits, duration limits, immediate feedback, and results.

Specify whether results are individual, aggregated by class, or both.

Documentation for this area is partial, as the screenshots mainly show the viewing of classes, presentations, and assessments, rather than administration workflows.

14. End-to-end workflows

Workflow A — Scan and study notes

Open the application and select a subject (or create one).

Tap the central scan button.

Capture the page and wait for OCR processing.

Review and correct the extracted text.

Select a subject and chapter, then confirm.

Choose to view the scanned content, request an explanation, generate content, or create a summary.

Use the content to create flashcards, a quiz, or a mind map.

Save the materials and return to them later.

Workflow B — Study with guidance from AiA

Open AiA with a subject or content item already selected.

The assistant presents the context and quick suggestions.

The student chooses a quiz, flashcards, an explanation, a summary, or an open-ended question.

The AI generates a response based on the available sources.

The student interacts, receives feedback, and can generate a new activity.

Workflow C — Check an essay

Open the Essay Checker.

Select the topic.

Take a photo or upload a PDF.

Submit and wait for processing.

Review the score, feedback, strengths, and recommendations.

Workflow D — Study through a learning path

Open the subject/topic learning path.

View the current activity and the next activity.

Complete the flashcards, quiz, or another activity.

Receive a status update and unlock the next stage.

15. Data, context, and proposed functional architecture

The following logical structure is a proposal to guide implementation; it is not a confirmed description of the existing technical architecture.

| Logical object | Responsibility |
|---|---|
| User/Profile | Identity, date of birth, country, language, and preferences. |
| LearningProfile | Goals, self-reported difficulties, and learning preferences. |
| Subject | The student's subjects. |
| Chapter | Content units within a subject. |
| SourcePage | Captured image, metadata, and extracted/ reviewed text. |
| GeneratedResource | Summary, explanation, quiz, flashcards, or mind map. |
| Attempt/Answer | Answers, feedback, score, and activity history. |
| StudyPath/Step | Sequence, status, unlocking, and progress. |
| Class/Presentation/Assessment | Structures for the teacher and class experience. |

Recommended technical principles

Persist processing requests and the status of each OCR/AI stage so that processing can resume after failures.

Use idempotency identifiers to prevent the same resource from being generated twice when requests are repeated.

Store both the OCR text version and the version reviewed by the user.

Record the model, version, prompt/template, and sources used for each generation to ensure traceability.

Implement clear states: pending, processing, completed, failed, and eligible for retry.

Enforce access control by user, class, and role.

Define privacy, retention, and deletion policies for images, text, and minors' data.

16. Non-functional requirements

| Dimension | Recommendation |
|---|---|
| Usability | Consistent interfaces, loading states, actionable error messages, and accessibility. |
| Performance | Immediate feedback when OCR/AI starts; asynchronous processing for long-running tasks. |
| Reliability | Controlled retries, state persistence, and duplicate prevention. |
| AI quality | Source-grounded responses, uncertainty indicators, and continuous evaluation. |
| Security | Authentication, authorisation, file protection, and data segregation by user/class. |

| | |
|---|---|
| Privacy | Data minimisation, consent, and policies appropriate for minors and the markets served. |
| Localisation | Consistent translations and country-specific curriculum adaptation, without conflating language and curriculum. |
| Observability | Logs for processing, OCR errors, generation, latency, and user feedback. |

17. Key considerations and priority recommendations

| Priority | Issue | Recommended action |
|---|---|---|
| P0 | Contradictory quiz feedback | Test answer selection, answer key, visual state, and explanation; fix the logic before scaling. |
| P0 | OCR accuracy | Implement quality validation and allow corrections before generating materials. |
| P0 | AI response reliability | Ground generated content in source material and flag uncertainty or missing information. |
| P1 | Inconsistent localisation | Standardise language and regional variants across all screens. |
| P1 | Scan limits | Formalise counting rules, notebook eligibility, and backend/interface synchronisation. |
| P1 | Progression and locked stages | Define explicit unlocking rules and learning path states. |
| P1 | Essay scoring | Explain the rubric, criteria, estimated nature of the score, and limitations of automated assessment. |
| P2 | Incomplete teacher area | Document class, student, material, assessment, and results management. |
| P2 | History and traceability | Allow users to locate the original source of each generated resource and resume sessions. |

18. Cross-functional acceptance criteria

The student can complete onboarding and later change the preferences that are editable.

The student can create and organise subjects and link chapters to them.

A captured page is processed, reviewed, and saved without losing the original image.

The reviewed text is used as the source for all subsequent generations.

All four post-scan actions work and retain a reference to the source page.

Flashcards and quizzes display accurate progress and content consistent with the source.

Quiz feedback always matches the selected answer and the answer key.

The mind map maintains a clear and coherent hierarchy.

AiA uses the correct context and does not mix subjects or users.

The essay checker provides an explainable assessment, and the semantic checker explains every highlighted passage.

Learning path states correctly reflect completion, the next activity, and locked stages.

The classes area displays content only to authorised users.

19. Open questions to finalise the specification

Is the application intended only for students, or also for teachers and parents/guardians?

Is the learning path automatic, manual, or hybrid?

Are the three free scans counted per account, device, time period, or installation?

How is a compatible Jandaia notebook validated?

Can generated materials be edited, exported, and shared?

Does AiA retain memory across subjects and sessions? What context limits apply?

What criteria and scoring scales does the essay checker use for each country/exam?

Does the semantic checker analyse only notes, or also answers and free-form text?

How are classes, presentations, and assessments created and managed?

Which progress and performance metrics are displayed to students and teachers?

20. Conclusion

The screenshots reveal a product with an integrated value proposition: starting with content actually created by the student, digitising it, organising it, and turning it into AI-supported learning experiences. The differentiating element is not merely the generation of quizzes or summaries, but the connection between the original page, the academic context, the AiA assistant, and practice and assessment activities.

To turn this experience into a development-ready specification, the next steps are to formalise the business rules, states and permissions, data contracts, AI quality criteria, and the teacher-area workflows that have not yet been demonstrated. The contradictory quiz feedback issue requires immediate validation.

**Final note: This report is a functional analysis based on the visual evidence provided and should be validated with the product and engineering teams before being treated as a contractual specification.**